SMARTCELL – Week 1 Activity Report

Week 1: Literature Study, Disease Research and Dataset Analysis

Objective of Week 1

The objective of Week 1 was to study the microscopic characteristics of the two selected diseases, **Malaria and Acute Lymphoblastic Leukemia (ALL)**, understand their blood-smear examination requirements, identify suitable public datasets, and collect information required for the development of the SMARTCELL prototype.

Activity 1 – Study of Malaria Microscopy

Source

World Health Organization (WHO) – Global Malaria Programme: Microscopy

Source: https://www.who.int/teams/global-malaria-programme/case-management/diagnosis/microscopy

Overview

Malaria is commonly detected by examining blood samples under a microscope. Two types of blood films are mainly used: **thick films**, which are useful for detecting parasites, and **thin films**, which help identify *Plasmodium* species and examine parasite morphology. **Giemsa staining** is commonly used to make the parasites visible under the microscope.

Microscopy can provide information about the presence of malaria parasites, their species and developmental stages, and can also help estimate parasite density. Accurate microscopic examination requires proper sample preparation, quality control, and trained personnel.

Data Collected

Parameter	**Finding**
Detection method	Blood smear microscopy
Blood film types	Thick and thin films
Common stain	Giemsa
Target organism	*Plasmodium* parasites
Information obtained	Parasite presence, species and stages

Additional measurement	Parasite density
Requirement	Proper preparation and trained examination

Outcome

The study provided the basic medical and microscopy requirements for malaria image analysis and helped identify the type of microscopic information that can be used in SMARTCELL.

Activity 2 – Study of Malaria Dataset

Source

NIH/NLM Malaria Dataset – Thin Blood Smears Pf

Source: https://data.lhncbc.nlm.nih.gov/public/Malaria/NIH-NLM-ThinBloodSmearsPf/ReadMe.pdf

Dataset Analysis

The NIH/NLM dataset was studied to understand the characteristics of microscopic blood images that can be used for computer-vision-based malaria research.

Dataset Parameter	**Collected Data**
Patients	193
Images per patient	5
Image resolution	5312 × 2988 pixels
Image channels	3
Smear type	Thin blood smear
Staining	Giemsa
Disease	*P. falciparum*
Healthy controls	Included
Annotation	Expert annotated
Image acquisition	Smartphone + microscope

Outcome

The dataset study provided information about the available malaria images, their resolution, staining, annotations and acquisition method. This information will be useful for dataset preparation and later image-processing and model-development stages.

Activity 3 – Study of Malaria Blood Smears

Source

Centers for Disease Control and Prevention (CDC) – Malaria Diagnostic Tests

Source: https://www.cdc.gov/malaria/hcp/diagnosis-testing/malaria-diagnostic-tests.htm

https://lhncbc.nlm.nih.gov/LHC-research/LHC-projects/image-processing/malaria-datasheet.html

Comparison of Blood Smears

Thick Blood Smear	**Thin Blood Smear**
Useful for parasite detection	Useful for species identification
Examines a larger amount of blood	Individual cells are easier to observe
Higher sensitivity for detecting parasites	Helps examine parasite morphology
Useful when parasite levels are low	Helps estimate parasitemia
More difficult to examine	Easier to observe individual cell structures

Outcome

The comparison helped identify the different purposes of thick and thin blood films and provided a basis for selecting appropriate microscopic image data for malaria-related image analysis.

Activity 4 – Study of Acute Lymphoblastic Leukemia (ALL)

Source

Diagnosis and Subclassification of Acute Lymphoblastic Leukemia – NCBI/PMC

Source: https://pmc.ncbi.nlm.nih.gov/articles/4235437/

Overview

Acute Lymphoblastic Leukemia (ALL) is a hematological malignancy involving lymphoid precursor cells. Microscopic examination can identify abnormal **lymphoblasts** in blood or bone-marrow samples. Morphological examination can provide useful information about blast-cell appearance, although additional laboratory techniques are required for definitive diagnosis.

Important Morphological Information

Parameter	**Finding**
Target cells	Lymphoblasts
Cell type	Immature lymphoid cells
Cell size	Variable, commonly small to intermediate
Nucleus	Generally round or oval
Nuclear-to-cytoplasmic ratio	Often high
Chromatin	Fine/dispersed in many lymphoblasts
Cytoplasm	Usually relatively scant
Microscopy role	Identification of abnormal blast-cell morphology

These features can help define which visual characteristics may be relevant during image-processing and classification research.

Outcome

The study established the basic microscopic characteristics of ALL and identified **lymphoblast detection and morphological analysis** as important image-analysis tasks for the SMARTCELL prototype.

Activity 5 – Study of ALL Image Dataset

Source

ALL-IDB – Acute Lymphoblastic Leukemia Image Database

Source: https://scotti.di.unimi.it/all/

Dataset Analysis

ALL-IDB is a public microscopic blood-image dataset specifically developed for evaluating **image segmentation, image classification and image-processing algorithms** for ALL. The dataset provides expert information about the location/classification of probable lymphoblasts.

ALL-IDB1

Dataset Parameter	**Collected Data**
Dataset	ALL-IDB1
Images	108
Approx. blood elements	39,000
Image format	JPG
Resolution	2592 × 1944 pixels
Color depth	24-bit
Magnification	Approximately 300–500×
Annotation	Expert oncologists
Target	Probable ALL lymphoblasts
Annotation information	Blast-cell centroid coordinates

The dataset can be used to study **image classification, blast-cell counting and blast-cell location detection**.

ALL-IDB2

Dataset Parameter	**Collected Data**
Dataset	ALL-IDB2
Image type	Cropped cell images
Classes	Normal / blast cell
Purpose	Cell classification
Image format	JPG
Source	ALL-IDB1

ALL-IDB2 contains cropped regions representing normal cells and probable blast cells, making it useful for cell-level classification research.

Outcome

The ALL-IDB datasets were identified as suitable public resources for studying **ALL cell detection, segmentation and classification**. The expert annotations can also be used for evaluating computer-vision algorithms.

Activity 6 – Comparison of Malaria and ALL Image Analysis

Parameter	**Malaria**	**ALL**
Disease	Malaria	Acute Lymphoblastic Leukemia
Main target	*Plasmodium* parasites / infected cells	Lymphoblasts
Sample	Blood smear	Blood/bone-marrow smear images
Common imaging approach	Microscopy	Microscopy
Important visual task	Parasite detection	Blast-cell detection
Dataset studied	NIH/NLM Malaria Dataset	ALL-IDB
Image processing relevance	Segmentation and parasite detection	Segmentation and cell detection
Classification task	Infected / non-infected or parasite-related classification	Normal / blast-cell classification

Outcome

The comparison established that SMARTCELL will require an image-processing pipeline capable of handling microscopic blood images and extracting disease-relevant cell or parasite regions before AI-based classification.

Activity 7 – Initial Image-Processing Requirements

Based on the malaria and ALL datasets, the following image-processing requirements were identified:

1. Image acquisition from a microscope or USB microscope.
2. Image resizing and standardization.
3. Noise reduction.
4. Contrast and illumination correction where required.
5. Segmentation of relevant regions.
6. Thresholding for separating foreground and background.
7. Cell or parasite detection.
8. Extraction of relevant regions.
9. Preparation of processed images for AI classification.

Proposed Pipeline

Microscope / USB Microscope

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Blood Image

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Image Preprocessing

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Segmentation

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Thresholding

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Cell / Parasite Detection

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Region Extraction

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AI Classification

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Preliminary Screening Output

Overall Week 1 Findings

During Week 1, the following activities were completed:

1. Studied malaria microscopy and blood-smear examination.
2. Studied thick and thin blood films.
3. Studied Giemsa staining and parasite detection.
4. Studied the NIH/NLM malaria dataset.
5. Recorded malaria dataset parameters and image characteristics.
6. Studied Acute Lymphoblastic Leukemia and lymphoblast morphology.
7. Studied the ALL-IDB image dataset.
8. Recorded ALL-IDB1 and ALL-IDB2 dataset characteristics.
9. Compared the image-analysis requirements of malaria and ALL.
10. Identified initial image-processing requirements for SMARTCELL.

Week 1 Final Outcome

The outcome of Week 1 was a **literature, microscopy and dataset research report covering both selected diseases: Malaria and Acute Lymphoblastic Leukemia (ALL).**

The report contains:

- Malaria microscopy study
- ALL microscopy and morphology study
- Malaria dataset analysis
- ALL-IDB dataset analysis
- Thick vs. thin blood-smear comparison
- Disease-specific image characteristics
- Dataset parameters and annotations
- Initial image-processing requirements
- Preliminary SMARTCELL image-processing pipeline

- Reliable references for further development

The collected information will be used in the next stage for **dataset preparation, OpenCV-based preprocessing, segmentation, cell/parasite detection and development of the preliminary AI-assisted screening pipeline**.

References

1. World Health Organization (WHO), Global Malaria Programme – Microscopy https://www.who.int/teams/global-malaria-programme/case-management/diagnosis/microscopy
2. NIH/NLM, Thin Blood Smears Pf Dataset https://data.lhncbc.nlm.nih.gov/public/Malaria/NIH-NLM-ThinBloodSmearsPf/ReadMe.pdf
3. Centers for Disease Control and Prevention (CDC), Malaria Diagnostic Tests https://www.cdc.gov/malaria/hcp/diagnosis-testing/malaria-diagnostic-tests.htm
4. ALL-IDB – Acute Lymphoblastic Leukemia Image Database https://scotti.di.unimi.it/all/
5. Diagnosis and Subclassification of Acute Lymphoblastic Leukemia – NCBI/PMC https://pmc.ncbi.nlm.nih.gov/articles/4235437/
6. ALL-IDB: The Acute Lymphoblastic Leukemia Image Database for Image Processing https://air.unimi.it/handle/2434/168333

SMARTCELL – Member 2

Week 1 Activity Report

Role: Image Processing & Cell Detection Research

Week 1 Objective

The objective of Week 1 was to study the basic image-processing techniques required for SMARTCELL, including **OpenCV, image preprocessing, segmentation, thresholding and cell detection**. The research focused on understanding how microscopic blood-cell images can be processed before applying AI-based classification.

1. OpenCV Research

Source

OpenCV – Official Documentation

Source: https://docs.opencv.org/

Research Performed

OpenCV is an open-source computer vision library that provides functions for image processing and computer-vision applications.

The following OpenCV operations were studied:

- Image reading and saving
- Image resizing
- Color-space conversion
- Grayscale conversion
- Image filtering
- Thresholding
- Contour detection

- Morphological operations
- Image segmentation
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Outcome

The basic OpenCV functions required for the SMARTCELL image-processing pipeline were identified.

Proposed pipeline:

Microscope Image

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OpenCV

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Image Preprocessing

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Segmentation

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Cell Detection

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Feature/Region Extraction

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AI Classification

2. Image Segmentation Research

Research Area

Image Segmentation in Microscopic Images

Segmentation is the process of separating the important regions of an image from the background or from other objects.

For SMARTCELL, segmentation can be used to separate:

- Blood cells
- Background

- Possible abnormal regions
- Individual cell regions

Techniques Studied

Technique	Purpose
Thresholding	Separate foreground from background
Contour detection	Identify object boundaries
Morphological operations	Remove noise and improve regions
Color-based segmentation	Separate regions based on pixel color
Connected components	Identify separate objects

Outcome

Segmentation was identified as an important preprocessing step for isolating individual blood-cell regions before further analysis.

3. Thresholding Research

Source

OpenCV Documentation – Thresholding

Source: https://docs.opencv.org/4.x/d7/d4d/tutorial_py_thresholding.html

Research Performed

Thresholding converts an image into different regions based on pixel intensity.

The following techniques were studied:

Technique	Basic Purpose
Binary Thresholding	Separates pixels using a fixed threshold
Binary Inverse	Produces the opposite binary result
Otsu Thresholding	Automatically determines a threshold
Adaptive Thresholding	Uses different thresholds for different image regions

Example Concept

Original Microscopic Image

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Grayscale

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Thresholding

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Binary Image

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Cell / Background

Outcome

Different thresholding techniques were identified for separating cell regions from the background. **Otsu and adaptive thresholding** were identified as techniques that can be tested on microscopic images.

4. Cell Detection Research

Research Area

Cell Detection using Computer Vision

Cell detection involves identifying the location and boundaries of individual cells within a microscopic image.

The following OpenCV techniques were studied:

- Contour detection
- Connected-component analysis
- Blob detection
- Morphological operations
- Bounding-box extraction

Basic Cell Detection Pipeline

Microscopic Image

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Preprocessing

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Grayscale

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Thresholding

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Morphological Processing

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Contour Detection

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Individual Cell Regions

Outcome

A basic computer-vision workflow for detecting individual blood-cell regions was identified. The detected regions can later be passed to the AI model for classification.

5. Image Preprocessing Research

Before segmentation and cell detection, microscopic images may require preprocessing to improve image quality.

Techniques Studied

Technique	**Purpose**
Resizing	Standardize image dimensions
Grayscale conversion	Simplify image information
Gaussian Blur	Reduce noise
Median Blur	Reduce noise while preserving edges
Histogram processing	Improve image contrast
Morphological operations	Improve segmented regions

Outcome

A preliminary preprocessing pipeline was defined:

Input Image

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Resize

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Noise Reduction

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Color/Grayscale Conversion

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Contrast Improvement

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Segmentation

6. Week 1 Practical Research Outcome

The research resulted in the following proposed image-processing pipeline for SMARTCELL:

USB Microscope

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Microscopic Blood Image

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OpenCV

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Image Preprocessing

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Grayscale / Color Processing

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Thresholding

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Segmentation

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Morphological Operations

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Cell Detection

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Individual Cell Regions

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AI Model

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Preliminary Screening Result

7. Week 1 Deliverables

The following outputs were prepared during Week 1:

1. OpenCV research notes
2. Image preprocessing technique list
3. Segmentation technique comparison
4. Thresholding technique comparison
5. Cell detection workflow
6. Proposed SMARTCELL image-processing pipeline
7. OpenCV documentation references
8. Initial technical requirements for implementation

8. Sources

OpenCV Official Documentation

https://docs.opencv.org/

OpenCV – Image Thresholding

https://docs.opencv.org/4.x/d7/d4d/tutorial_py_thresholding.html

OpenCV – Contours

https://docs.opencv.org/4.x/d4/d73/tutorial_py_contours_begin.html

OpenCV – Morphological Transformations

https://docs.opencv.org/4.x/d9/d61/tutorial_py_morphological_ops.html

OpenCV – Connected Components

https://docs.opencv.org/4.x/d3/dc0/group__imgproc__shape.html

Final Week 1 Outcome – Member 2

Member 2 completed the initial research on OpenCV-based image processing, segmentation, thresholding and cell detection. A preliminary image-processing pipeline was designed to convert microscopic blood images into separated cell regions that can later be analyzed by the AI model.